

electron/graph/extract-structure.mjs

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Overview

Adds a new CLI script `electron/graph/extract-structure.mjs` that deterministically extracts structural data from a batch of files using TreeSitter and non-code parsers, replacing earlier regex-based logic.

Key changes

- **Imports:** Adds Node core imports (`createRequire` , `dirname` , `resolve` , `join` , `fileURLToPath` , `pathToFileURL` , `readFileSync` , `writeFileSync`) at lines 19-22.
- **Dynamic core loading:** Resolves `@novadiff/graph-core` with `pathToFileURL` to support Windows ESM paths (lines 35-41).
- **CLI entry point:** Implements `async function main()` (lines 48-140) that parses `process.argv` , reads input JSON, processes each file, builds results via `buildResult` , and writes output JSON.
- **Result builder:** Exports `buildResult(file, totalLines, nonEmptyLines, analysis, callGraph, batchImportData)` (lines 147-305) to map analysis data to the expected output schema.
- **CLI guard:** Detects direct execution and runs `main()` (lines 311-321).

Impact

- **Correctness:** Provides deterministic extraction using TreeSitter and non-code parsers; replaces fragile regex scripts.
- **Maintainability:** Centralizes result construction in a pure function (`buildResult`), making it unit-testable and easier to modify.
- **Performance:** Skips analysis for files >12,000 lines, reducing runtime on large codebases.
- **Compatibility:** Uses `pathToFileURL` to handle Windows paths, ensuring ESM imports work cross-platform.
- **Observability:** Emits detailed metrics (`importCount` , `exportCount` , etc.) and logs errors to `stderr` .

Risks & follow-ups

- **Path resolution:** Verify `pathToFileURL` handling on Windows; ensure `require.resolve('@novadiff/graph-core')` resolves correctly in all environments.
- **Batch import data:** Confirm `batchImportData` is populated; fallback logic may miscount imports if data is missing.
- **Large file handling:** Files >12,000 lines are skipped from analysis; ensure this threshold matches project expectations.

- **CLI argument parsing:** Test with missing or extra arguments to confirm error messages and exit codes.